

UCT722: INTRODUCTION TO IOT

L	T	P	Cr
2	1	2	3.5

Course objectives: This course will help students understand basic principles and concepts of Internet-of-Things use cases, applications, architecture and technologies. Students will get an overview of an end to end IoT system encompassing the edge, cloud and application tiers. This course will build upon the foundations created in the pre-requisite courses and will equip the students to architect a complete IoT application on their own. The lab exercises will consist of hands-on experiments that will lead to building an IoT application end-to-end. Some of the specialized topics will be covered via student seminars where students are expected to research and present their findings in a seminar format.

UNIT – I

Introduction to IoT and Use cases: Understanding basic concepts of IoT, Consumer IoT vs Industrial Internet, Fundamental building blocks, Use Cases of IoT in various industry domains.

UNIT – II

Architecture: IoT reference architectures, Industrial Internet Reference Architecture, Edge Computing, IoT Gateways, Data Ingestion and Data Processing Pipelines, Data Stream Processing.

UNIT – III

Sensors and Industrial Systems: Introduction to sensors and transducers, integrating sensors to sensor processing boards, introduction to industrial data acquisition systems, industrial control systems and their functions.

UNIT – IV

Networking and Communication for IoT: Recap of OSI 7 layer architecture and mapping to IoT architecture, Introduction to proximity networking technologies (ZigBee, Bluetooth, Serial Communication), Industrial network protocols (Modbus, CANbus), Communicating with cloud applications (web services, REST, TCP/IP and UDP/IP sockets, MQTT, Web Sockets, protocols. Message encoding (JSON, Protocol Buffers).

UNIT – V

IoT Data Processing and Storage: Time Series Data and their characteristics, time series databases, basic time series analytics, data summarization and sketching, dealing with noisy and missing data, anomaly and outlier detection,

IoT Seminars:

Selected topics in IoT should be handled via student seminars. Recommended that students form a group do research on at least one of the following topics and present it through

seminars. They are expected to do a literature survey of the topic and present their survey paper to the class. The suggested topics are –

a) IoT Applications

- Smart Cities
- Connected Vehicles and Telematics
- Smart Grids
- Smart Homes

b) IoT data visualization

c) Survey of cloud based IoT platforms

d) Low power wide area networks for IoT

e) IoT device management

f) Survey of chips, embedded modules and development boards for IoT devices

g) Embedded and real-time operating systems for IoT

h) IoT Security

- Security risks in IoT
- Securing IoT endpoint devices and secure communication protocols for IoT
- Security and Privacy of IoT data

Lab Exercises

1. Setting up the Arduino Development Environment, connecting analog sensors to an Arduino Boarding and reading analog sensor data.
2. Digital Input and Output reading using and Arduino board and Arduino Development Environment.
3. Integrate an Arduino Board to a Raspberry Pi computer and send sensor data from Arduino to the R Pi.
4. Setup Python on the R Pi and run sample R Pi programs on the R Pi. Read the data from Arduino using Python language.
5. Connect a R Pi Camera module to the Raspberry Pi and using Python programming capture still images and video.
6. Set up TCP/IP socket server on a PC. Send a message from the R Pi to the PC using socket communication.
7. Set up a MQTT broker on the PC. Send data from R Pi to PC using MQTT protocol. Receive data from PC to R Pi using MQTT protocol.
8. Connect LED lights to an Arduino. Connect the Arduino to the R Pi. Send Message from PC to R Pi via MQTT protocol. On receipt of the message, toggle the LED lights on the Arduino.
9. Set up an account in a cloud service (such as Google / AWS or Azure). Set up a simple Http server using a language of your choice. Push the image captured from the R Pi camera to this web service. On receiving the image, store the image in a database or file.
10. Develop a mobile application to view the images captured by the R Pi camera.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

Upon completion of this course, students will be able to:

1. To Test and design some IOT based prototypes.

2. Understand the working principles of different types of sensors.
3. To describe various layers of IoT protocols stack along with its functionalities.
4. To understand the privacy and security issues and their related industry based standard solutions.
5. To provide skill and knowledge for demonstrating and building IoT based projects with the help of coding, hardware devices, sensor requirements, emulating and testing.
6. To develop an understanding for mapping wireless communication protocols and networking concepts for Industrial IoT deployments.

Text Books:

1. The Internet of Things, Samuel Greengard, MIT Press Essential Knowledge Series.

Reference Books / Links:

1. Industrial Internet Reference Architecture - <http://www.iiconsortium.org/IIRA.htm>.
2. World Economic Forum Report on Industrial Internet of Things - <https://www.weforum.org/reports/industrial-internet-things>.
3. 50 Sensor Applications for a Smarter World - http://www.libelium.com/resources/top_50_iot_sensor_applications_ranking/.
4. Visualizing Data-Exploring and Explaining Data with the Processing Environment, By Ben Fry, Publisher: O'Reilly Media.
5. Raspberry Pi Computer Architecture Essentials, by Andrew K Dennis.
6. Getting Started with Arduino, M. Banzi, O Reilly Media.
7. GSMA IoT Security Guidelines & Assessment - <https://www.gsma.com/iot/future-iot-networks/iot-security-guidelines/>.